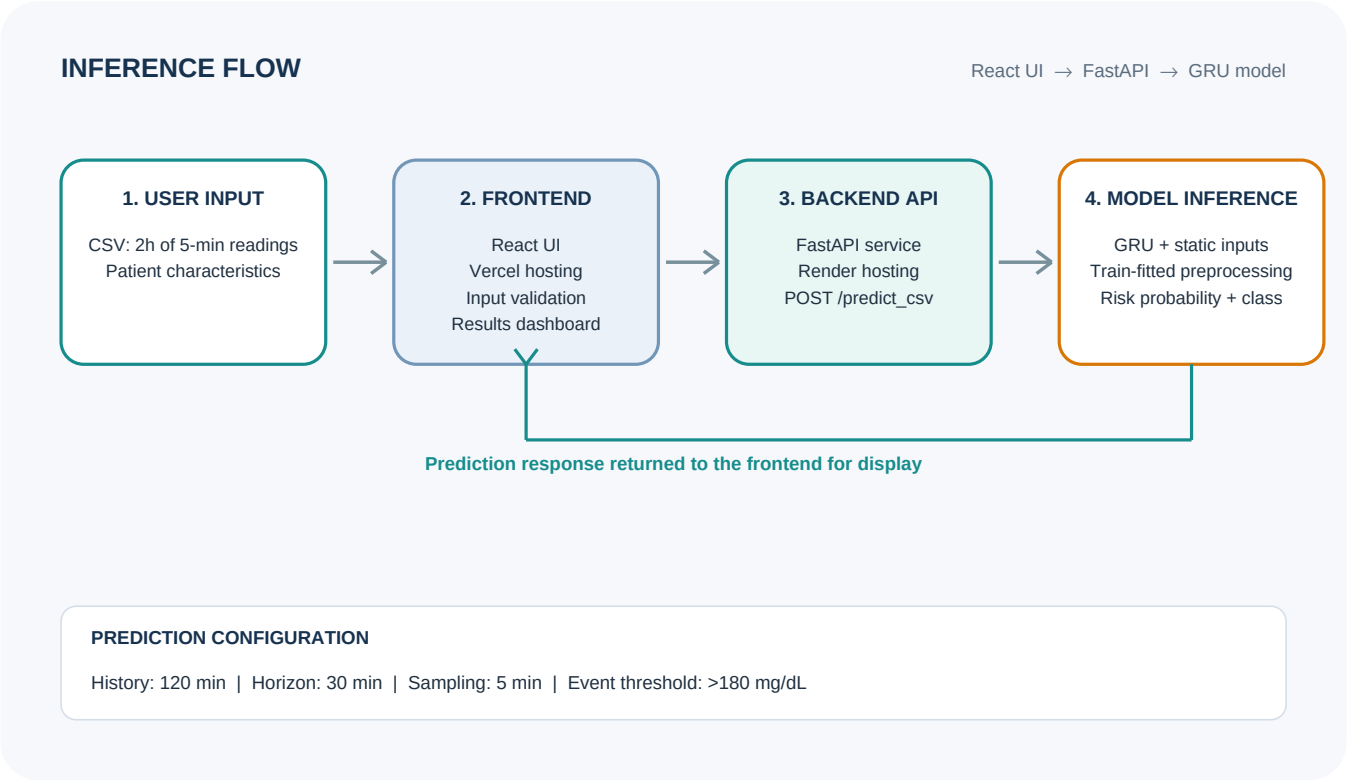
System Architecture

Meet Your Digital Twin: Personalized Glucose Risk Prediction

This document summarizes the deployed prototype's request flow, inference components, and prediction configuration.



Request & Prediction Flow

1. Input	The user supplies a compatible CSV containing the previous two hours of physiological readings and provides patient characteristics in the web interface.
2. Frontend	The React application, deployed on Vercel, collects the inputs, checks the expected format, and presents the prediction results.
3. API	The FastAPI backend, deployed on Render, receives the request at `/predict_csv` and prepares the inputs for inference.
4. Inference	The GRU model uses the time-series history alongside encoded and scaled static characteristics to produce a probability and a threshold-based class.
5. Response	The backend returns the prediction response, and the frontend displays the estimate and glucose-history visualizations.

Prediction Configuration

Historical input	Previous 2 hours (25 readings at 5-minute intervals)
Prediction horizon	Next 30 minutes (6 future readings)

Event definition	Glucose strictly above 180 mg/dL for six consecutive readings
Selected prototype model	GRU with static patient characteristics
Decision threshold	0.7496, selected on validation data

■ Deployment Components

Frontend	React application	Vercel
Backend	FastAPI inference API	Render
Model	GRU + static patient characteristics	TensorFlow / Keras

■■ Scope & Disclaimer

This is an experimental educational prototype, not a clinically validated medical device. Its predictions must not be used for diagnosis, treatment decisions, insulin dosing, or as a replacement for glucose monitoring or professional medical advice.

Live application: https://digital-twin-glucose-risk-cef3z27v3-aahana-ml.vercel.app/ | API documentation: https://digital-twin-glucose-risk.onrender.com/docs